

GeN-Foam: a novel OpenFOAM® based multi-physics solver for 2D/3D transient analysis of nuclear reactors



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HIGHLIGHTS

- Development of a new multi-physics solver based on OpenFOAM®.
- Tight coupling of thermal-hydraulics, thermal-mechanics and neutronics.
- Combined use of traditional RANS and porous-medium models.
- Mesh for neutronics deformed according to the predicted displacement field.
- Use of three unstructured meshes, adaptive time step, parallel computing.

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ABSTRACT

The FAST group at the Paul Scherrer Institut has been developing a code system for reactor analysis for many years. For transient analysis, this code system is currently based on a state-of-the-art coupled TRACE-PARCS routine. This work presents an attempt to supplement the FAST code system with a novel solver characterized by tight coupling between the different equations, parallel computing capabilities, adaptive time-stepping and more accurate treatment of some of the phenomena involved in a reactor transient. The new solver is based on OpenFOAM®, an open-source C++ library for the solution of partial differential equations using finite-volume discretization. It couples together a multi-scale fine/coarse mesh sub-solver for thermal-hydraulics, a multi-group diffusion sub-solver for neutronics, a displacement-based sub-solver for thermal-mechanics and a finite-difference model for the temperature field in the fuel. It is targeted toward the analysis of pin-based reactors (e.g., liquid metal fast reactors or light water reactors) or homogeneous reactors (e.g., fast-spectrum molten salt reactors). This paper presents each “single-physics” sub-solver and the overall coupling strategy, using the sodium-cooled fast reactor as a test case, and essential code verification tests are described.

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1. Introduction

Simulation of the transient behavior of nuclear reactors has been based for many years on the loose coupling of available legacy codes for neutron diffusion and 1-D or sub-channel thermal hydraulics (Mikityuk et al., 2005; Kozłowski et al., 2014). Typically the coupling is achieved through data exchange interfaces that exchange the most relevant information at each time step. This explicit exchange of information does not usually consistently resolve the system non-linearities, resulting in inaccuracies or requiring the use of extremely short time steps in order to achieve stable and

accurate solutions (Mahadevan et al., 2012; Ragusa and Mahadevan, 2009). A second problem is that legacy codes have often been written decades ago and typically do not support parallel computing, thus limiting the size of problems that can be investigated. The last, perhaps most critical problem associated to the coupling of available codes relates to the difficulties of modification. This aspect is becoming crucial following the need for more accurate simulations (CASL, 2011), the growing interest in particular aspects of reactor analysis (e.g., the inter-assembly flow in sodium-cooled fast reactors – SFRs (Tenchine, 2010)), and the development of innovative reactor systems. This has pushed e.g., the molten salt reactor community toward the development of a new generation of solvers (Aufiero et al., 2014; Fiorina et al., 2014), which often features modern capabilities in terms of coupling, flexibility, and parallel computing. Similar efforts have been directed toward the

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analysis of very high temperature reactors (Clifford, 2013; Clifford et al., 2013). The need of accurate or more detailed simulations has also been favoring a modernization of the tools used by the LWR (light water reactors) industry, as witnessed by the CASL initiative (CASL, 2011) and the MOOSE project (Gaston et al., 2009) in the US, as well as by the NURESAFE project in EU (NURESAFE, 2013).

The FAST group at the Paul Scherrer Institut has been active for many years in the field of multi-physics simulation of FRs. Specifically, a coupled TRACE-PARCS routine has been adapted for fast reactor (FR) analysis and validated against relevant experimental data (Chenu et al., 2012). Following the trends discussed above, efforts are currently being directed toward stepping beyond explicit data exchange interfaces and use of legacy codes. The objective is to develop a modern multi-physics platform where: (1) the various physics can be solved in a more implicitly coupled manner; (2) the growing computational resources available can be used at their fullest through proper parallelization algorithms; and, (3) modifications can easily be made to the source code to adapt it to new needs. In view of the involvement of the FAST group in FR analysis, two other important objectives are an improved capability in terms of thermal-mechanical analysis and the capability to simulate inter-assembly flow.

An available numerical library has been used to reduce the development efforts while making use of the latest computational techniques in the field of engineering. Specifically, the C++ based library OpenFOAM® (OpenFOAM, 2014; Weller et al., 1998) has been selected. This library includes all necessary routines for the discretization and parallel solution of partial differential equations on unstructured meshes using finite-volume methods. It also includes solvers for thermal-hydraulics and thermal-mechanics analyses that can be used as a starting point for the development of a coupled solver for reactor analysis. Based on OpenFOAM®, a multi-physics solver has been developed that couples:

- A thermal-hydraulics sub-solver based on the standard $k-\varepsilon$ turbulence model for compressible or incompressible flows, but extended to coarse-mesh applications through the use of a porous medium approach (Vafai, 2005) for user-selected cell zones inside the mesh. This allows for instance to simulate a full primary circuit using standard CFD techniques for open spaces such as the plena in LWRs or the pools in SFRs, while employing a coarse mesh with dedicated sub-scale models in complex components like core and heat exchangers.
- A displacement-based thermal-mechanics sub-solver based on the solidDisplacementFoam solver already available in OpenFOAM®. Both fine and coarse mesh investigations are possible, provided that appropriate material properties are used.
- A multi-group neutron diffusion sub-solver with user-selected energy group structure and provided with a dedicated routine to export nuclear data from the Monte Carlo Serpent-2 code (Leppänen, 2007) to OpenFOAM®-readable input files. The sub-solver includes moving-mesh features, as the mesh is deformed based on the displacement field provided by the thermal-mechanics sub-solver.
- A finite-difference sub-scale fuel model that can be used in coarse-mesh simulations of the core to evaluate the local temperature profile in fuel and cladding.

A first-order implicit Euler scheme is used for time integration. The coupling between equations is semi-implicit using Picard iteration (Ragusa and Mahadevan, 2009). Three different meshes are used for thermal-hydraulics, thermal-mechanics and neutron diffusion, while the sub-scale fuel model is solved in each mesh cell within the fuel zones of the thermal-hydraulics mesh. This allows different refinement of the meshes and reduces computational requirements. Consistent mapping is performed between

the three meshes at each time step and information is passed between different meshes using a cell-volume-weighted algorithm (OpenFOAM, 2014). The multi-physics solver is inherently 3-D, but 2-D (Cartesian or axial-symmetric) simulations can be performed with appropriate geometry and boundary conditions. It can be used for pin-based (e.g., SFRs and LWRs) or liquid fueled (e.g., MSRs) reactor systems. Currently, only single-phase fluid flow is allowed.

Following the broad range of possible applications, the different level of details (coarse or fine mesh) and the possibility to selectively include the different “physics” to solve, the multi-physics solver has been named Generalized Nuclear Foam – GeN-Foam.

The objective of this paper is to concisely present the main features of GeN-Foam in terms of theoretical background and numerical approach. The application to an SFR design is used throughout the paper as illustration to offer an overview about the possible range of applications of GeN-Foam. The same reactor design is also used to provide an initial set of essential code verification tests.

The paper is organized as follows. Section 2 presents the single-physics sub-solvers employed in GeN-Foam, including examples of application and verification tests. Section 3 describes the coupling strategy. Section 4 presents a preliminary code-to-code benchmark with the TRACE code (Mikityuk et al., 2005; Spore et al., 2000). The conclusions of the work are drawn in Section 5 while Appendix A presents the SFR design as used in this work.

2. GeN-Foam single-physics sub-solvers

The GeN-Foam multi-physics solver is built upon 4 main components, namely: a thermal-hydraulics sub-solver; a thermal-mechanics sub-solver; a neutronics sub-solver; and a simple finite-difference 1-D sub-solver that evaluates temperatures in fuel and cladding in case the core is simulated using a coarse mesh. A common feature for all sub-solvers is the use of an unstructured mesh, which allows for a notable flexibility in terms of geometrical configurations, and distinguishes GeN-Foam from many available codes that operates exclusively on structured mesh or even on selected geometrical configurations (e.g., geometries composed by hexagonal or square assemblies). All sub-solvers have parallel computation capabilities and parallelization is achieved through geometrical domain decomposition. Discretization of the equations is performed according to a finite-volume approach (except for fuel temperatures) and solution of the resulting algebraic systems is achieved using built-in preconditioned linear iterative solvers in OpenFOAM®. The following subsections describe the four main sub-solver components, including some basic verification tests that have been performed. Details about the exact spatial discretization schemes and solution algorithms for each equation are not reported for brevity. Time integration uses a first-order implicit Euler scheme.

2.1. Thermal-hydraulics

A main objective of GeN-Foam is to investigate the full core or primary circuit of nuclear reactors. Standard turbulence modeling (e.g., Reynolds averaged Navier–Stokes (RANS)) is currently too demanding for routine calculations at core level, so that alternative approaches must be pursued. A standard engineering approach to investigate complex structures using a coarse mesh is to treat these structures as porous media, with dedicated models employed to simulate the interaction of the fluid with the sub-scale structures (Vafai, 2005). An advantage of this treatment is that the equations employed revert back to standard RANS equations in case of clear fluid, so that the same set of equations can be discretized and solved on the same mesh while de facto treating different zones of the

geometry with two different approaches (detailed RANS or coarse mesh porous medium). This alleviates the numerical problems (e.g., of stability or numerical diffusion) that can instead be encountered when trying to couple different solutions from different meshes or from different solvers, as in the frequent case of a coupling between 1-D codes for core analysis and 3-D codes for pool or plenum RANS calculations.

2.1.1. Governing equations

Equations for the turbulent single-phase flow of a fluid in a porous medium can be rigorously derived from standard Navier–Stokes equations via time and volume averages. The resulting equations of mass, momentum and energy conservation are (Clifford, 2013; Saurel and Abgrall, 1999; Vafai, 2005):

$$\frac{\partial \gamma \rho}{\partial t} + \nabla \cdot (\gamma \rho \mathbf{u}) = 0 \quad (1)$$

$$\frac{\partial \gamma \rho \mathbf{u}}{\partial t} + \nabla \cdot (\gamma \rho \mathbf{u} \otimes \mathbf{u}) = \nabla \cdot (\mu_T \nabla \mathbf{u}) - \nabla \gamma p + p_i \nabla \gamma + \gamma \mathbf{F}_g + \gamma \mathbf{F}_{ss} \quad (2)$$

$$\frac{\partial \gamma \rho e}{\partial t} + \nabla \cdot (\mathbf{u} \gamma (\rho e + p)) = \nabla \cdot (\gamma k_T \nabla T) + \gamma \mathbf{F}_{ss} \cdot \mathbf{u} + \gamma \dot{Q}_{ss} \quad (3)$$

It can be observed that Eqs. (1)–(3) closely resemble the traditional RANS equations (Patankar, 1980). A first main difference is the appearance of a porosity term γ , which takes into account the fact that only part of the volume is occupied by the fluid. Moreover, two additional terms $\mathbf{F}_{ss}$ and $\dot{Q}_{ss}$ enter the equations. These two terms represent the effect of the sub-scale structures on the fluid flow. In case of clear fluid these terms are equal to zero and the porosity equal to one, so that Eqs. (1)–(3) become the traditional RANS equations.

The porosity enters the spatial derivatives in Eqs. (1)–(3), which poses numerical issues at the interfaces between cell zones with different porosities. An additional problem is that the interfacial pressure p_i needs to be modeled. A standard assumption (Clifford, 2013; Saurel and Abgrall, 1999) is that p_i equals p , which changes Eq. (2) into:

$$\frac{\partial \gamma \rho \mathbf{u}}{\partial t} + \nabla \cdot (\gamma \rho \mathbf{u} \otimes \mathbf{u}) = \nabla \cdot (\mu_T \nabla \mathbf{u}) - \gamma \nabla p + \gamma \mathbf{F}_g + \gamma \mathbf{F}_{ss} \quad (4)$$

thus bringing the porosity outside of the gradient in the pressure term. To further limit the problem associated with a possibly discontinuous porosity, it is convenient to rewrite Eq. (4) in terms of the Darcy velocity defined as:

$$\mathbf{u}_D \triangleq \gamma \mathbf{u} \quad (5)$$

which is by definition continuous at the interface between cell zones with different porosities. Eqs. (1)–(3) can then be rewritten as:

$$\frac{\partial \gamma \rho}{\partial t} + \nabla \cdot (\rho \mathbf{u}_D) = 0 \quad (6)$$

$$\begin{aligned} \frac{\partial \rho \mathbf{u}_D}{\partial t} + \frac{1}{\gamma} \nabla \cdot (\rho \mathbf{u}_D \otimes \mathbf{u}_D) \\ = \nabla \cdot (\mu_T \nabla \mathbf{u}) - \gamma \nabla p + \gamma \mathbf{F}_g + \gamma \mathbf{F}_{ss} - (\rho \mathbf{u}_D \otimes \mathbf{u}_D) \nabla \frac{1}{\gamma} \end{aligned} \quad (7)$$

$$\begin{aligned} \frac{\partial \gamma \rho e}{\partial t} + \nabla \cdot (\mathbf{u}_D (\rho e + p)) \\ = \gamma \nabla \cdot (k_T \nabla T) + \mathbf{F}_{ss} \cdot \mathbf{u}_D + \gamma \dot{Q}_{ss} + (k_T \nabla T) \cdot \nabla \gamma \end{aligned} \quad (8)$$

Porosity now enters a spatial derivative only in the last terms in the right-hand sides of Eqs. (7) and (8). The last term in Eq. (8) can safely be neglected in most problems of interest in the field of nuclear engineering, especially in view of the fact that sudden changes of porosity typically happen in the direction of the flow, where the convective term is dominant over the diffusion term.

The last term of Eq. (7) represents instead the Bernoulli effect. As a matter of fact, in most applications in nuclear engineering, the porosity does not change inside a porous zone, but only at the interface between two different zones. The last term in Eq. (7) can thus be neglected (at least for low Mach number flows) provided its effect is properly taken into account at interfaces between different cell zones. This can be done in OpenFOAM® via introduction of a pressure baffle, i.e., a forced discontinuity in pressure. In fact such a pressure baffle is necessary to take into account localized pressure drops at the entrance of a porous zone. An alternative approach is to include a thin buffer region at the interface between two different porous zones, and to define there the term $\mathbf{F}_{ss}$ so that it models both the concentrated pressure drops and the Bernoulli effect.

Solution of Eqs. (6)–(8) is achieved through a merged PISO–SIMPLE (named PIMPLE) pressure-based algorithm for compressible flows (Issa, 1986; OpenFOAM, 2014; Patankar, 1980). The sub-solver allows for non-compressible solutions via simplification of the solution algorithm, with optional inclusion of a Boussinesq approximation to include buoyancy effects.

2.1.2. Sub-scale momentum sink

The term $\mathbf{F}_{ss}$ in Eq. (7) represents the drag force exerted by the sub-scale structures on the fluid. As such, it is proportional to velocity. Such proportionality is typically linear, quadratic or between the two. It is then convenient to express $\mathbf{F}_{ss}$ as:

$$\mathbf{F}_{ss} = \kappa(\mathbf{u}_D) \cdot \mathbf{u}_D \quad (9)$$

The convenience of Eq. (9) relies on the possibility to treat implicitly in the numerical solver the linear dependence of $\mathbf{F}_{ss}$ on $\mathbf{u}$. The tensor $\kappa(\mathbf{u}_D)$ can be obtained from experiments and is often expressed as the sum of a constant term plus a term linearly dependent on the velocity magnitude (Vafai, 2005). However, in many engineering applications such term has a dependence on velocity which is neither constant nor linear, so that no initial assumption has been made on its dependence in the sub-solver here presented.

The exact value and velocity dependence of $\kappa(\mathbf{u}_D)$ is typically not available. However, if selecting the coordinate system so that it is aligned with the principal coordinate axes of the porous zone, such tensor becomes diagonal. In addition, when using the principal coordinate system of a porous zone, correlations for pressure drops are typically available for the three principal directions and can be used to estimate the three diagonal components of $\kappa(\mathbf{u}_D)$. In fact, by considering $\kappa(\mathbf{u}_D)$ as diagonal and assuming the flow as one-directional in one of the three principal directions, one can easily derive from Eqs. (7) and (9):

$$(\nabla p)_i = F_{ss,i} = \kappa(u_D)_{ii} u_{D,i} \quad (10)$$

∇p can be estimated (Incropera et al., 2006) as:

$$(\nabla p)_i = \frac{f_{D,i} \rho u_{D,i}^2}{2 D_h \gamma^2} \quad (11)$$

where f_D and D_h are the Darcy friction factor and hydraulic diameter (or characteristic length) of the sub-scale structure for the considered principal direction. Eqs. (10) and (11) finally yield:

$$\kappa(u_D)_{ii} = \frac{f_{D,i} \rho u_{D,i}}{2 D_h \gamma^2} \quad (12)$$

which allows to derive the tensor $\kappa(\mathbf{u}_D)$ from available correlations or data for pressure drops. The term f_D is assumed in the sub-solver to have the form:

$$f_{D,i} = A_{f_{D,i}} Re^{B_{f_{D,i}}} \quad (13)$$

The constants D_h , $A_{f_{D,i}}$ and $B_{f_{D,i}}$ are user selected and the Reynolds number Re is calculated using the velocity magnitude $|\mathbf{u}|$ and the hydraulic diameter D_h . To fully define $\kappa(\mathbf{u}_D)$ in its principal coordinate system, one has then to provide three different values for $A_{f_{D,i}}$ and $B_{f_{D,i}}$. In fact, two sets of parameters can be used for laminar and turbulent flow conditions. In addition, the user must provide the two Reynolds numbers marking the end of a laminar regime and the beginning of a fully turbulent regime, so that six values of $A_{f_{D,i}}$ and $B_{f_{D,i}}$ must be provided. In the transition region, the sub-solver uses a Reynolds-weighted arithmetic mean between the values of $f_{D,i}$ for laminar and fully turbulent flow (RELAP5-3D, 2005).

It is worth noting that the global coordinate system in a simulation is often not coincident with the principal coordinate system of all the porous zones contained in the domain. For this reason, the possibility has been included in the sub-solver to specify the principal coordinate system of each porous zone. The sub-solver will then rotate the velocity in the porous zone, evaluate $\mathbf{F}_{ss}$ in the rotated (local) coordinate system (for which correlations in the form (13) are assumed to exist) and then rotate back $\mathbf{F}_{ss}$ to the global coordinate system before use in Eq. (7).

2.1.3. Sub-scale energy sink/source

The heat source Q_{ss} in Eq. (8) represents the volumetric heat which is transferred between the fluid and the sub-scale structure and can be expressed as:

$$\dot{Q}_{ss} = A_V h (T_{ss} - T) \quad (14)$$

with the local temperature of the sub-scale structure T_{ss} governed by the following equation:

$$\rho_{ss} c_{p,ss} \frac{\partial T_{ss}}{\partial t} = \nabla \cdot (\gamma \mathbf{k}_{ss} \nabla T) + A_V h (T - T_{ss}) \quad (15)$$

The heat transfer coefficient h is evaluated following an approach similar to that used for $\kappa(\mathbf{u}_D)$. Specifically, for each principal direction of a porous medium, the Nusselt number is evaluated according to correlations of the form:

$$Nu_i = A_{Nu,i} Re^{B_{Nu,i}} Pr^{C_{Nu,i}} + D_{Nu,i} \quad (16)$$

And the heat transfer coefficients h in Eqs. (14) and (15) is evaluated as

$$h = \sum_i \frac{h_i u_{D,i}}{|\mathbf{u}_D|} \quad (17)$$

A set of parameters for Eq. (16) is given for each principal direction and for both laminar and turbulent flow conditions. Similarly to the case of the Darcy friction factor, the sub-solver uses an Reynolds-weighted arithmetic mean between the values of h_i for laminar and fully turbulent flow. To allow for a simplified treatment of heat exchangers, the possibility is included to model the heat exchange with an external fluid at fixed temperature by modifying Eq. (14) as:

$$\dot{Q}_{ss} = A_V h_{eff} (T_{ext} - T) \quad (18)$$

where h_{eff} is calculated as the harmonic mean of h and a user-selected external heat transfer coefficient.

2.1.4. Turbulence modeling

The terms μ_T and k_T in Eqs. (7) and (8) are modeled in clear fluid zone using the standard k - ε model for RANS equations (Launder

and Sharma, 1974). For porous zones, the problem of predicting k and ε becomes extremely complex. Some approaches have been proposed in the literature (Chandesris et al., 2006; Nakayama and Kuwahara, 1999; Pedras and de Lemos, 2001a,b,c; Teruel, 2007; Teruel and Rizwan-uddin, 2009) but their applicability to general engineering cases remains limited. As a matter of fact, pressure drops and heat transfer in porous zones are typically dominated by the interaction with the sub-scale structure, so that an accurate modeling of μ_T and k_T becomes inessential. On the other hand, it is relatively important to predict a correct value of k and ε at the entrance of a clear fluid zone, which asks for a good prediction of k and ε at the outlet of the porous zone where the fluid comes from (this can be the case of the upper plenum in a LWR or the pool of an SFR when modeling the core as a porous medium).

Following these considerations, GeN-Foam avoids solving the standard k - ε model inside the porous zones, but instead forces the values of k and ε to converge to user-selected values (k_0 and ε_0) with a user-selected convergence rate $\lambda_{\varepsilon/k}$ by solving the following equations:

$$\rho \frac{\partial \varepsilon}{\partial t} + \nabla \cdot (\rho \mathbf{u}_D \varepsilon) = \rho \lambda_{\varepsilon/k} (\varepsilon_0 - \varepsilon) \quad (19)$$

$$\rho \frac{\partial k}{\partial t} + \nabla \cdot (\rho \mathbf{u}_D k) = \rho \lambda_{\varepsilon/k} (k_0 - k) \quad (20)$$

The same set of equations is solved for both clear fluid and porous zones. In particular, Eqs. (19) and (20) are derived from the standard k - ε equations employed in the clear-fluid zones via elimination of selected terms and introduction of new ones.

Similarly to the case of Darcy friction factor and Nusselt number, three different values of k_0 and ε_0 must be used, one for each principal direction of the porous zone. The values to be used in Eqs. (19) and (20) are then derived similarly to Eq. (17).

If no numerical or experimental information is available to evaluate k_0 and ε_0 at the outlet of a porous zone, a possible approximate solution is to use their equilibrium value inside the porous zone while neglecting their local variation in the outlet region. To evaluate the equilibrium value of k and ε in a porous zone it is often possible to compute the turbulent length scale and the turbulent intensity from available correlations of the form:

$$l = A_l Re^{B_l} \quad (21)$$

$$l = A_l D_h \quad (22)$$

The turbulent length scale and the turbulent intensity can then be used to evaluate ε and k as (Launder and Sharma, 1974):

$$I = \frac{1}{|\mathbf{u}|} \left(\frac{2}{3} k \right)^{0.5} \quad (23)$$

$$l = 0.1643 \frac{k^{3/2}}{\varepsilon} \quad (24)$$

2.1.5. Application to a SFR

The thermal-hydraulics of a SFR can be simulated using GeN-Foam according to different degrees of approximation. As an example, Fig. 1 shows a 2-D r-z model of the European sodium fast reactor (ESFR, Appendix A) pool, with all components treated as porous zones except for the hot and cold pools, where the use of a standard turbulence model allows to predict regions of recirculation and stagnation of the flow (Fig. 1b). Internal boundaries (baffles) are used to separate hot and cold pool. For each porous zone, a complete set of parameters must be supplied to the code to describe the interaction with the sub-scale structure. As an example, Table 1 shows the set of parameters used for the cell zone representing the core. It should be noted that no indication of local coordinate axes is given to the sub-solver because the local principal axes correspond to the global coordinate system. In addition,

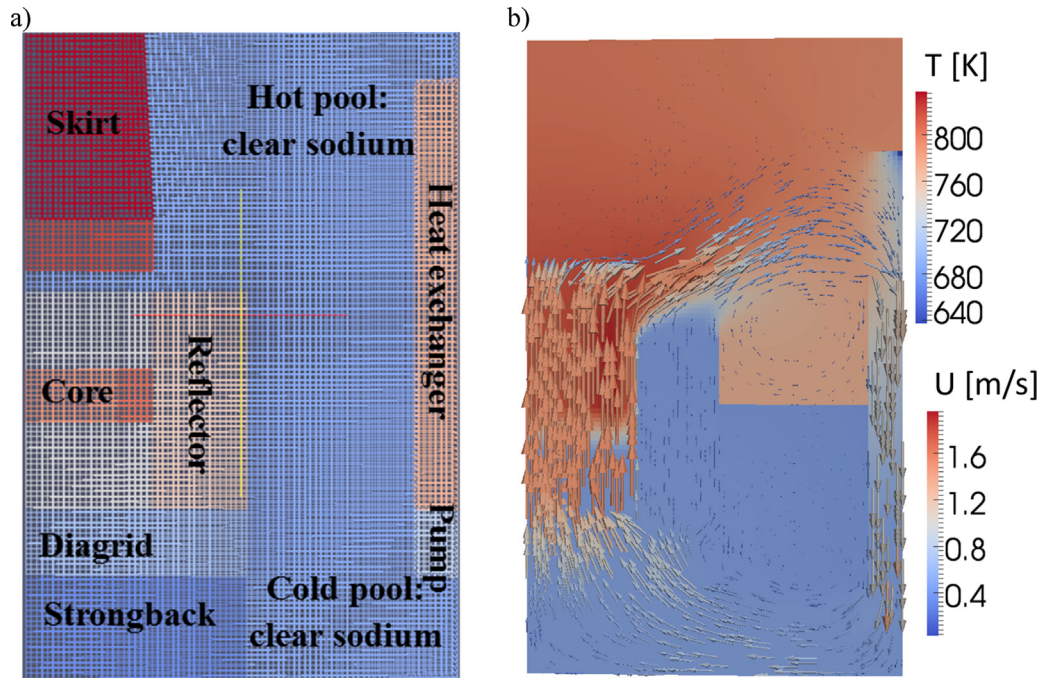


Fig. 1. 2-D thermal-hydraulic analysis of the ESFR pool at operating conditions: (a) geometry and mesh with cell zones; (b) sodium temperatures and velocities.

Table 1
Set of parameters used to describe the interaction of the fluid with the fuel bundle.

Parameter	Value	Source of information
γ	0.395	ESFR geometrical description
A_V	237.4 m^{-1}	ESFR geometrical description
Transition Reynolds to laminar	1000	(Incropera et al., 2006) ^a
Transition Reynolds to turbulent	2300	(Incropera et al., 2006) ^a
$A_{f,D,i}$ (turbulent)	0.687	(Bubelis and Schikorr, 2008)
$B_{f,D,i}$ (turbulent)	−0.25	(Bubelis and Schikorr, 2008)
$A_{f,D,i}$ (laminar)	64	(Incropera et al., 2006) ^a
$B_{f,D,i}$ (laminar)	−1	(Incropera et al., 2006) ^a
D_h	3.65 mm	ESFR geometrical description
$A_{Nu,i}$ (turbulent)	0.018	(Mikityuk, 2009)
$B_{Nu,i}$ (turbulent)	0.8	(Mikityuk, 2009)
$C_{Nu,i}$ (turbulent)	0.8	(Mikityuk, 2009)
$D_{Nu,i}$ (turbulent)	4.6	(Mikityuk, 2009)
$A_{Nu,i}$ (laminar)	0	(Incropera et al., 2006) ^a
$B_{Nu,i}$ (laminar)	0	(Incropera et al., 2006) ^a
$C_{Nu,i}$ (laminar)	0	(Incropera et al., 2006) ^a
$D_{Nu,i}$ (laminar)	4.36	(Incropera et al., 2006) ^a
A_l	0.16	(Pope, 2003) ^a
B_l	−1/8	(Pope, 2003) ^a
C_l	0.07	(Pope, 2003) ^a

^a Without considering the wire spacer.

one single value is given for each of the parameters in Eqs. (13) and (16) following the 1-D nature of the flow in the core. In fact, a high value of $A_{f,D,i}$ could be used for the directions perpendicular to the flow motion to limit the radial flow in the core, which would be unphysical in the ESFR, where the fuel elements are enclosed in a wrapper.

As a second example of application, Fig. 2 shows a 3-D simulation of the ESFR core. In this case, baffles are used to separate the flow in different assemblies. In view of the coarse mesh employed, a slip boundary condition for velocity must be used on these boundaries to avoid any coolant cross-flow over these boundaries without affecting the axial flow. The assumption behind this is that

pressure drops are dominated by the interaction with the sub-scale structure (the fuel bundle in this case), making the simulation of the velocity gradient close to the assembly wall unnecessary.

Finally, Fig. 3 shows a detail of another 3-D simulation of the ESFR core. Compared to the previous case, the inter-assembly gap is explicitly modeled in this case. The gap between assemblies is only 4.5 mm thick and the small mesh element size necessary to simulate it can pose severe Courant number constraints on the time step. To limit this problem the gap has been treated as a porous medium by including the wrapper as part of it. This generates a gap of 13.5 mm with a ~33% porosity. In case of explicit simulation of the inter-assembly flow, an interesting possibility offered by OpenFOAM® relates to the simulation of conduction across a baffle (representing the wrapper in this case). In fact, it is possible to specify on both sides of a baffle a dedicated boundary condition (named thermal baffle) that simulates heat conduction in the structure that the zero-thickness baffle is in fact representing. This is done via generation of an additional 3-D mesh where standard time dependent heat conduction equations are solved and implicitly coupled with the temperature solution in the original mesh.

It is worth mentioning that the use of a coarse mesh to simulate the heat transfer between assembly and inter-assembly gap requires the fluid temperature gradient at wall (not simulated if using a coarse mesh) to be small compared to the gradient in the wrapper. This assumption has been preliminary verified through a fine-mesh simulation, showing that neglecting the temperature gradient in the fluid may lead to a 5–10% underestimation of the heat transfer (Bonet, 2014). Although this might be considered as acceptable, it is worth noting that the use of an appropriate convective (Robin) boundary condition at the wrapper may strongly reduce such approximation.

2.1.6. Code verification

The thermal-hydraulic sub-solver presented in the previous paragraphs has been derived from an available OpenFOAM® solver (compressiblePimpleFoam (OpenFOAM, 2014)). A comprehensive set of verification tests has been performed to ascertain the

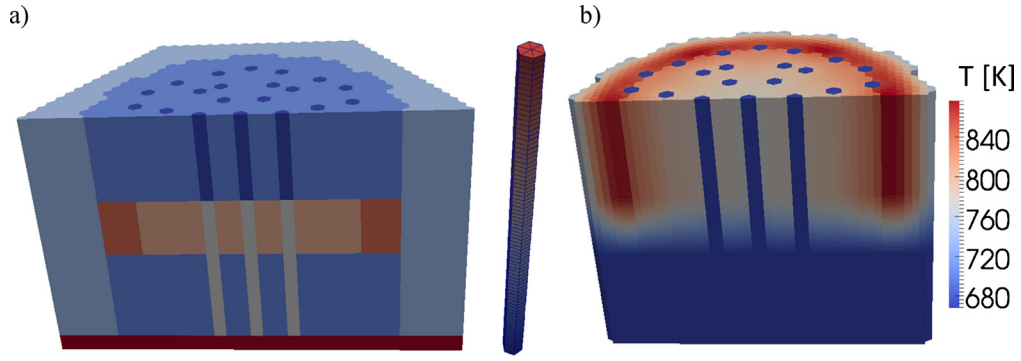


Fig. 2. 3-D thermal-hydraulic analysis of the ESFR core at operating conditions: (a) geometry with cell zones and mesh detail for one assembly; (b) sodium temperatures (without reflector).

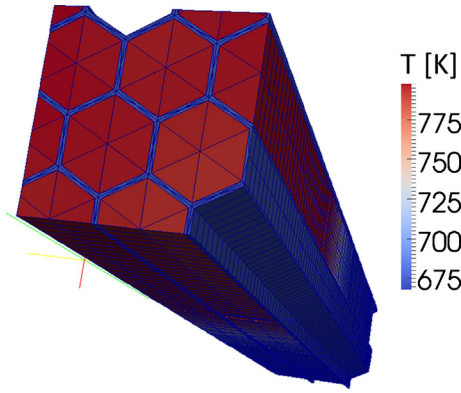


Fig. 3. Detail of a 3-D thermal-hydraulic analysis of the ESFR core at operating conditions, including simulation of the inter-assembly sodium flow.

correct implementation of each modification included in the original solver. These tests include:

- Comparison with analytic results for 1-D (x direction) simulations of porous structures where each one of the parameters in Eqs. (13), (16), (21), (22) has been sequentially modified.
- Comparison with analytic results for 1-D simulations using a local coordinate system for the porous structure that is rotated compared to the global coordinate system.
- Consistency check based on natural convection cavity flow using both the compressible version of the sub-solver and the incompressible version with Boussinesq approximation.

All tests were successful and more details about this verification work can be found in Bonet (2014).

2.2. Sub-scale fuel

Eq. (15) is used in GeN-Foam to evaluate the temperature of the sub-scale structure in all cell zones treated as porous media. However such equation is not adequate for zones containing the fuel, where an accurate prediction of the temperature profile in fuel and cladding is necessary. Eq. (15) is then substituted with a more complex model. Axial-symmetric geometry is assumed, axial heat conduction in the fuel is neglected and, for each cell in the fuel, a 1-D heat conduction problem is solved for pellet and cladding:

$$\rho_f C_{p,f} \frac{\partial T_f}{\partial t} = k_f \frac{\partial^2 T_f}{\partial r^2} + k_f \frac{1}{r} \frac{\partial T_f}{\partial r} + \dot{Q}_f \quad (25)$$

$$\rho_c C_{p,c} \frac{\partial T_c}{\partial t} = k_c \frac{\partial^2 T_c}{\partial r^2} + k_c \frac{1}{r} \frac{\partial T_c}{\partial r} \quad (26)$$

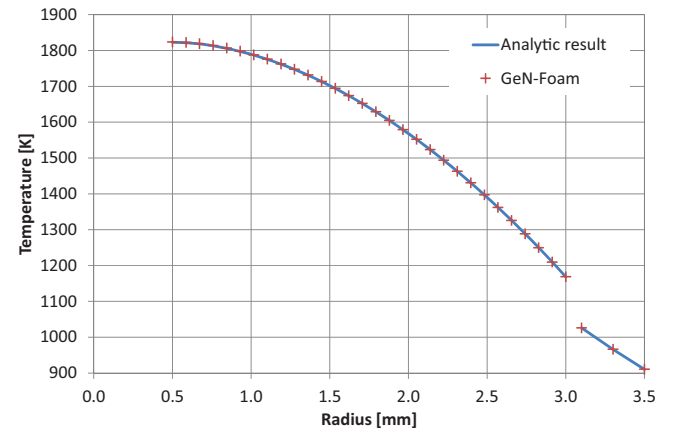


Fig. 4. Example of comparison between numerical and analytic results for the temperature field in pellet and cladding.

Eqs. (25) and (26) are discretized using a first-order implicit Euler scheme in time and a second-order finite-difference scheme in space (see Ozisik (1994) for details). The coupling with the solution for coolant temperature provided by Eq. (8) is achieved by calculating the heat flux on the clad surface and transferring it to the coolant in the form of a volumetric heat source via multiplication with the fuel volumetric area. Conversely, the coolant temperature and the heat transfer coefficient between coolant and cladding are used to set up a convective boundary condition for Eq. (26). An analogous procedure is used to couple Eqs. (25) and (26) through definition of a gap conductance and convective (Robin) boundary conditions. The sub-solver includes the possibilities of simulating both solid and annular fuel pins and the number of nodes in fuel and cladding is user selectable.

Verification of the sub-scale fuel temperature model has been performed against analytic results, showing a very good agreement for a number of radial nodes equal to or higher than 10 in the fuel and 3 in the cladding. As an example, Fig. 4 shows the comparison between analytic and numerical results for an annular fuel pin using 30 nodes in the fuel and 3 in the cladding.

2.3. Thermal-mechanics

A thermal-mechanic solver named solidDisplacementFoam available in the OpenFOAM® 2.3.0 version has been used as basis for the solution of the thermal-mechanics equations. It is a displacement-based transient segregated finite-volume solver of linear-elastic, small-strain deformation of a solid isotropic body, with optional thermal diffusion and thermal stresses. The equations

employed by the solver are not reported here for brevity, but details can be found in (OpenFOAM, 2014).

Although the solver envisions the possibility of heterogeneous media, its original version required the thermo-mechanical properties to be supplied to the solver in the form of fields. For convenience, the possibility has been included to use meshes subdivided into multiple homogenous cell zones, and to associate a set of properties to each cell zone through an input file. This has been the only significant modification made to the solver and no verification tests were performed, other than some successful sanity checks where the solutions have been compared to reference solutions obtained using the original solver.

The thermo-mechanic solver described above is used as a sub-solver of GeN-Foam to evaluate the temperature-induced deformation of the main structures such as vessel, strongback, and diagrid, the latter determining the radial displacement in the core. Axial deformation of the neutronic mesh in the active core is calculated independently, based on the fuel (in case of open gap) or cladding (in case of closed gap) expansion coefficients and temperatures. Specifically, an equation is solved in the form:

$$\mathbf{v}_f \cdot \nabla D_f = \alpha_{f/c}(T_{f/c} - T_{f/c,ref}) \quad (27)$$

where $\mathbf{v}_f$ is the user-selected axial orientation of the fuel. An equation similar to Eq. (27) is also used to determine the axial displacement of the control rod driveline. The three displacement fields are then combined to deform the neutronic mesh. The temperatures of sub-scale structures, fuel, cladding and control rod drivelines are calculated by the thermal-hydraulic sub-solver (Section 2.1) and mapped into the thermal-mechanics mesh using a cell-volume-weighted technique.

As an example, Fig. 5 shows the displacement fields at operating conditions (zero-displacement corresponds to room temperature) for the ESRF core. All structures have been modeled using a simplified geometry that reproduces their main contours while neglecting their details (according to a preliminary coarse-mesh philosophy, see Fig. 5a). In addition, only the diagrid and the radial reflector have been modeled as rigid structures, while the rest is modeled as a soft medium with a Young modulus two orders of magnitudes lower than diagrid and radial reflector. This is an artifice to correctly predict the mesh deformation in the core caused by the expansion of the diagrid (Fig. 5b). Figs. 5c and 5d display the axial displacement of fuel and control rod drivelines as calculated from equations in the form of Eq. (27). Finally, Fig. 5e shows the displacement field obtained by combining the displacements in Figs. 5b, 5c and 5d. The latter is used to deform the neutronic mesh. Masses in the core are adapted to the mesh deformation based on appropriate parameterization of cross-sections (see next Section).

2.4. Neutronics

Solution of the spatial neutron kinetics is based on the traditional multi-group diffusion equations:

$$\begin{aligned} \frac{1}{v_i} \frac{\partial \varphi_i}{\partial t} = & \nabla \cdot D \nabla \varphi_i + \frac{v \Sigma_{f,j}(1 - \beta_{eff,t}) \chi_i}{k_{eff}} \varphi_i \\ & - \sum_{d,i} \varphi_i + \frac{S_{n,i}(1 - \beta_{eff,t}) \chi_i}{k_{eff}} + S_d \chi_i + S_{s,i} \end{aligned} \quad (28)$$

where the explicit source terms $S_{n,i}$, $S_{d,i}$ and $S_{s,i}$ are calculated as:

$$S_{n,i} = \sum_{j \neq 1} v \Sigma_{f,j} \varphi_j \quad (29)$$

$$S_d = \sum_k \lambda_k C_k \quad (30)$$

$$S_{s,i} = \sum_{j \neq 1} \Sigma_{j \rightarrow i} \varphi_j \quad (31)$$

and resolved via Picard iterations. It should be noted that $\beta_{eff,t}$ is used in Eq. (28), with the same neutron yield for both prompt and delayed neutrons. As a matter of fact the sub-solver also allows distinguishing between prompt and delayed neutron yields and using $\beta_{0,t}$ instead of $\beta_{eff,t}$. The precursor concentrations C_k in Eq. (30) are evaluated using the following equations for the delayed neutron precursor production and decay (and convection in case of liquid circulating fuel):

$$\frac{\partial C_k}{\partial t} + \nabla \cdot (\mathbf{u}_D C_k) = \frac{\beta_{eff,k} \sum_j v \Sigma_{f,j} \varphi_j}{k_{eff}} - \lambda_k C_k \quad (32)$$

The sub-solver is capable of time dependent or eigenvalue calculations. In the latter case, a traditional power iteration algorithm (Stacey, 2007) is used and the prompt neutron production term of the i th group in Eq. (28) is left explicit. The number and structure of both neutron energy and precursors groups is arbitrary.

As mentioned, the mesh for neutronics calculations is deformed according to the displacement field calculated by the thermal-mechanic sub-solver. Specifically, the displacement field is projected from the thermal-mechanics to the neutronics mesh, interpolated from the cell centers (where it is calculated) to the cell vertices and used to displace the mesh points.

Similarly to the thermal-mechanics and thermal-hydraulics, the mesh for neutronics can be subdivided into cell zones and a cross-section set can be associated to each zone. In fact, different cross-section sets for different perturbed states can be used in each zone. Using cell-wise information of the perturbed variable, the sub-solver evaluates the local cross-sections to be used by interpolating between nominal and perturbed cross-section sets. The interpolation can be linear or not. For instance, for the ESRF, six cross-sections set are supplied for each cell zone, namely: (1) nominal; (2) increased fuel temperature; (3) axially expanded fuel; (4) radially expanded core; (5) reduced coolant density; (6) expanded cladding. At each time step or power iteration, the correct cross-section set is evaluated in each cell based on local temperatures of fuel, cladding and coolant, as well as local radial and axial expansions. The interpolation is linear in all cases but the fuel temperature, for which a logarithmic interpolation is known to be more suited for FRs (Waltar et al., 2012). Fig. 6 shows for the ESRF an example of geometry and mesh (Fig. 6a), and solution for the flux in both an un-deformed (Fig. 6b) and deformed geometry (Fig. 6c). It should be noted that OpenFOAM® offers discretization schemes up to fourth order in space, which has been used in the case of the ESRF to achieve a mesh-independent solution by using only 24 cells for the radial discretization of each assembly (as in Fig. 5a for thermal-mechanics). This corresponds to a relatively large characteristic length of approximately 10 cm, to be compared with an average mean free path of ~3 cm in the active core region.

To generate the sets of cross-sections employed by the sub-solver, the Monte Carlo code Serpent-2 can be used and both a Matlab (Matlab, 2005) and an Octave (Eaton et al., 2008) routines have been implemented to automatically transfer the cross-section into a format suitable for use in OpenFOAM®. For the case of the ESRF, six full core simulations (based on the Serpent input from Ghasabyan (2013)) have been performed to generate the cross-sections in each zone, adopting homogenization methods recently implemented in Serpent-2 (Fridman and Leppänen, 2012).

Main code verification tests (not reported here for the sake of brevity) based on analytic solutions have been run to test the capability of the sub-solver to reproduce k_{eff} and zero-power transient behavior of neutrons and precursors in an infinite geometry and in

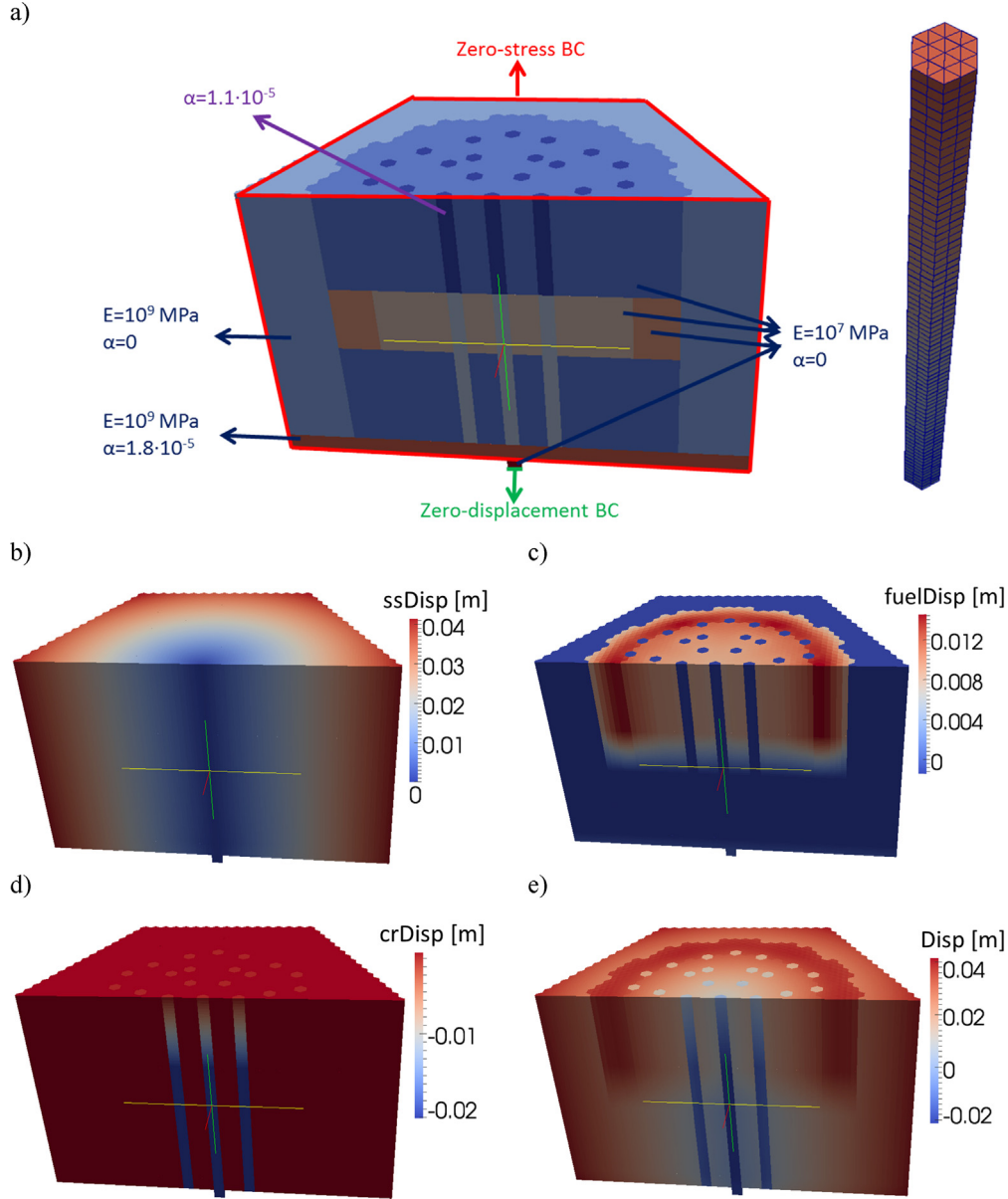


Fig. 5. Thermal-mechanic analysis of the ESRF core at operating conditions: (a) geometry with cell zones, assumptions and mesh detail for one assembly; (b) radial displacement of the structures; (c) axial displacement of the active core; (d) axial displacement of the control rods; (e) combined displacement field used to deform the neutronics mesh.

a homogeneous cylinder. A second verification step has also been carried out by comparing the results with 1 energy group of GeN-Foam with those provided by Serpent-2. The k_{eff} for the nominal core state has first been evaluated. The capability of the sub-solver to reproduce a perturbed core state has then been investigated and predicted reactivity variations have been compared to Serpent-2 results. Results are listed in Table 2. A generally good agreement can be observed between the results provided by Serpent-2 and GeN-Foam. In particular, the k_{eff} is slightly overestimated (~ 400 pcm) and feedback coefficients are reproduced with good accuracy. A significant discrepancy ($\sim 20\%$) can be observed only for the case of the coolant density effect. This is consistent with the use of a diffusion solver, as coolant density reduction causes a strong change in diffusion coefficient in the control rod follower, which is mainly composed by sodium. Calculations with 33 energy groups have also been performed for the nominal core state, resulting in a k_{eff} equal to 1.01557, i.e., ~ 500 pcm lower compared to Serpent-2. The latter result is consistent with results obtained with the ERANOS

diffusion and transport solvers (Rimpault et al., 2002) for a very similar core configuration (Pelloni et al., 2011).

3. Coupling strategy

The GeN-Foam multi-physics solver is built upon the main four sub-solver components described in Sections 2.1–2.4. Fields (solution variables) calculated on one mesh but needed on a second mesh are projected based on a standard OpenFOAM® cell-volume-weighted algorithm. The coupling strategy is depicted with some simplifications in Fig. 7, with specific reference to the compressible case. The simulation starts by selecting a time step. Initial time step is given by the user while subsequent time steps are decided based on the most stringent requirement between Courant number condition in the fluid and a maximum allowed power variation, which is set by default to 2.5% but can be selected by the user.

After selection of the time step, GeN-Foam follows the structure of the PIMPLE loop for solving velocity, pressure and energy.

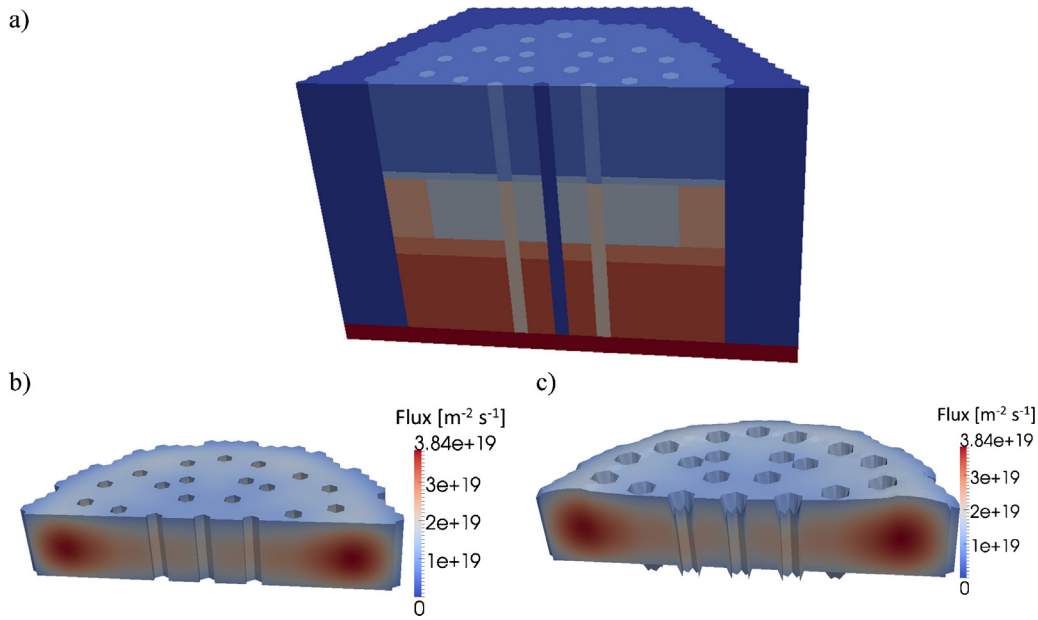


Fig. 6. Neutronic analysis of the ESFR core at operating conditions: (a) geometry; (b) solution in the un-deformed mesh (active core only); (c) solution in the deformed mesh (active core only, displacement magnified by 20 times).

Table 2

Comparison between GeN-Foam and Serpent-2 results for the ESFR neutronics.

	Reactivity difference (pcm)					
	k_{eff} (–)	Coolant density reduced to 500 kg/m ³	5% fuel expansion	5% radial core expansion	Fuel temperature increased by 300 K	5% cladding expansion
Serpent-2	1.02047	423	–1333	–2037	–343	498
GeN-Foam	1.02464	339	–1284	–1990	–334	551

Specifically, density equation (i.e., mass conservation, Eq. (6)) is first solved, followed by solution of k and ε equations. An optional velocity predictor step is then performed based on Eq. (7) and using the pressure field from the previous iteration. A pressure correction equation is then solved sequentially a specified number of times to correct for non-orthogonality effects. The pressure equation is a Poisson equation that can be obtained through manipulation of Eqs. (6) and (7) (see e.g., Patankar (1980) for details). Using the new pressure field, an explicit form of Eq. (7) is then used to correct the velocity field. This predictor-corrector loop is repeated until the desired degree of convergence is obtained. Under-relaxation can be used while iterating and it is automatically excluded at the last iteration. After achieving the velocity field, the energy equation Eq. (8) is solved and a number of “outer correctors” are performed to resolve the coupling of pressure, velocity and energy.

After achieving the coolant flow and temperature fields, GeN-Foam solves for the temperatures in sub-scale structures, fuel and cladding (Eqs. (15), (25) and (26)) and again for the energy equation (Eq. (8)). It then goes through the thermal-mechanic sub-solver and the neutronic sub-solver, both including their own Picard iterations to solve for the explicit terms. Initial residuals are evaluated before solving each equation. They are calculated based on a normalized difference between the right-hand and the left-hand sides of the discretized form of an equation. The equation is not solved if these residuals are lower than a user-selected tolerance. If all initial residuals are found to be lower than the desired tolerance, GeN-Foam moves to the next time step. In case one of the residuals is found to be higher, it restarts from calculating temperatures in fuel and sub-scale structures.

Iteration between pressure-velocity-energy solution and the other equations is typically not necessary and such possibility has not been shown in Fig. 7. However, the possibility of a user-selected number of “outer iterations” steps is included in GeN-Foam. Finally it is worth mentioning that the possibility has been included to solve only for selected equations, while leaving the other fields unaltered.

Computational requirements for the simulations performed have been: (1) several minutes up to 1 h on single core for a 2-D combined coarse-fine mesh application (Fig. 1); (2) few hours up to one day on ~10 cores for a 3-D coarse mesh simulation (Figs. 2, 5 and 6). Preliminary scalability tests have shown promising performances, as expected by making standard use of OpenFOAM® routines.

4. Preliminary code-to-code benchmark with trace

A code-to-code benchmark with the TRACE system code has been performed in order to verify the correct implementation of the coupling algorithm described in Section 3, as well as to provide an additional verification for the single-physics sub-solvers presented in Sections 2.1–2.4. To this purpose, a 3-D GeN-Foam model has been built using the geometries and meshes shown in Figs. 2, 5 and 6, data from Appendix A, and one-group cross-sections calculated using the Serpent model presented by Ghasabyan (2013) (and modified for use in Serpent-2). For simplicity, a radially flat velocity profile has been considered. The TRACE model consists of a single fuel rod cell featuring the same average power and coolant velocity as the GeN-Foam model. Neutronics is solved in TRACE using its built-in point-kinetics model, with generation time and

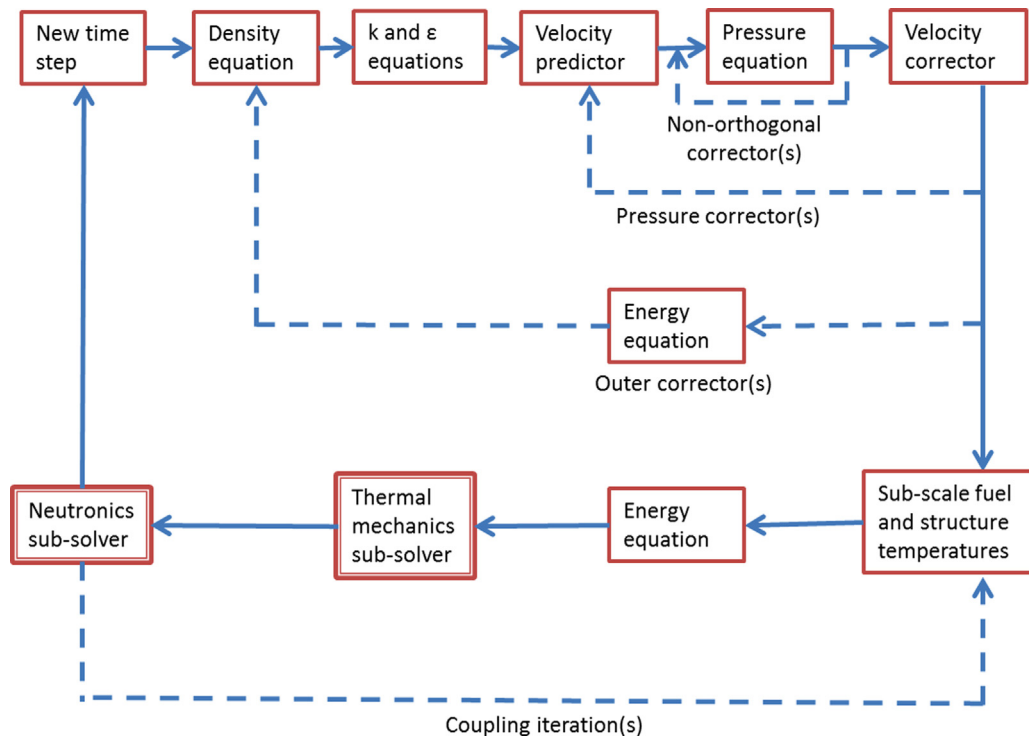


Fig. 7. GeN-Foam coupling strategy.

Table 3

Neutron kinetics parameters used for the TRACE point kinetic model.

Generation time	0.507 μ s
Beta effective/precursor decay constant	
Group 1	5.8 pcm/0.0125 s ⁻¹
Group 2	61.8 pcm/0.0283 s ⁻¹
Group 3	22.5 pcm/0.0425 s ⁻¹
Group 4	56.4 pcm/0.133 s ⁻¹
Group 5	12.3 pcm/0.292 s ⁻¹
Group 6	54.6 pcm/0.666 s ⁻¹
Group 7	45.1 pcm/1.63 s ⁻¹
Group 8	22.8 pcm/3.55 s ⁻¹
Doppler constant	952 pcm
Fuel expansion coefficient	-0.328 pcm/K
Coolant expansion coefficient	
Active core	0.503 pcm/K
Above active core	-0.0825 pcm/K

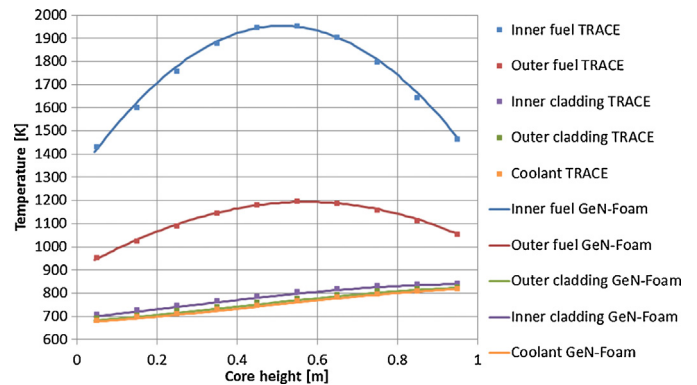


Fig. 8. Comparison of GeN-Foam and TRACE results in terms of average axial temperature profiles.

Beta effective from adjoint-weighted Serpent-2 calculations, based on the Iterated Fission Probability method (Leppänen et al., 2014), and the feedback coefficients obtained through eigenvalue calculations with GeN-Foam (Table 3). Axial power density shape has been taken from the Serpent-2 model. Radial expansion and cladding expansion have not been modeled in TRACE and the respective cross-section dependencies have thus been switch off in GeN-Foam.

4.1. Steady-state results

As a first step for the code verification phase, steady-state results provided by the two codes have been compared. Specifically, temperature profiles have been compared in different radial locations of the average pin, namely: coolant (bulk temperature), outer surface of the cladding, inner surface of the cladding, outer surface of the fuel, and inner surface of the fuel. Obtained results are reported in Fig. 8, which shows that no significant discrepancies exist between the results of the two codes.

4.2. Transient results

To test the capability of GeN-Foam to correctly predict a reactor transient behavior, two different accidental transients have been simulated and results compared with TRACE.

As a first benchmark case, an Unprotected Transient OverPower (UTOP) has been simulated where 100 pcm reactivity has been inserted linearly in 2 s. Such reactivity insertion has been achieved via modification of the k_{eff} and of the number of neutrons produced per fission in TRACE and GeN-Foam, respectively. Fig. 9 shows the resulting power, average coolant temperature and average fuel temperature as predicted by the two codes. GeN-Foam predicts a lower peak power and slightly higher asymptotic power and average temperatures. However, a generally good agreement is observed. In fact, discrepancies of this magnitude can typically be expected between a point-kinetics and a spatial-kinetics model (Chenu et al., 2012).

As a second test case, an Unprotected Loss Of Flow (ULOF) has been simulated through an exponential decay of the flow rate to

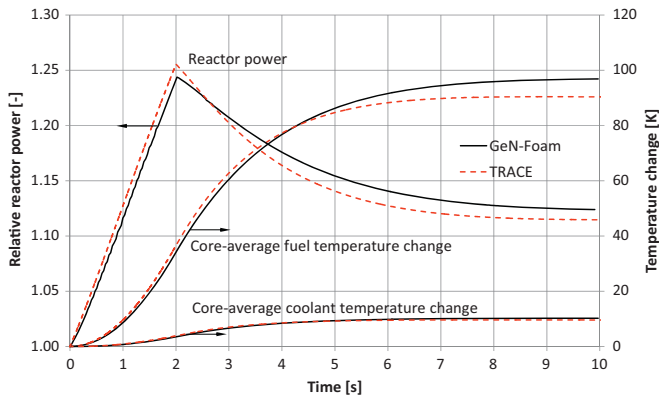


Fig. 9. Comparison of GeN-Foam and TRACE results for an UTOP in terms of power, average fuel temperature, and average coolant temperature.

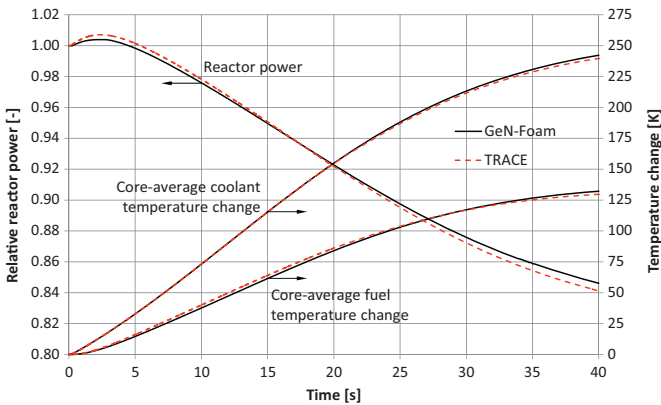


Fig. 10. Comparison of GeN-Foam and TRACE results for an ULOF in terms of power, average fuel temperature, and average coolant temperature.

20% of the nominal value with a halving time of the flow rate of 10 s. Fig. 10 shows the resulting power, average coolant temperature and average fuel temperature as predicted by the two codes. A very close agreement is observed between the two solutions.

5. Conclusions and future developments

A new multi-physics solver for nuclear reactor analysis, named GeN-Foam, has been presented. It is based on OpenFOAM® and has been developed for the multi-physics transient analysis of pin-based (e.g., liquid metal FRs, LWRs) or homogeneous (e.g., fast spectrum MSRs) nuclear reactors. It includes the solution of coarse or fine mesh thermal-hydraulics, thermal-mechanics and neutron diffusion. In particular, the thermal-hydraulics solution combines both a traditional RANS model and a porous medium model on the same mesh, depending on the desired degree of approximation for each zone. If the active reactor core is modeled as a porous medium, a simple sub-solver has been prepared for computing the sub-scale radial temperature profiles in fuel and cladding. The mesh used for neutronics calculations is deformed according to the displacement field predicted by the thermal-mechanics solver, thus allowing for a direct prediction of expansion-related feedback effects in FRs. To reduce computational requirements and increase flexibility, GeN-Foam permits the use of three different unstructured meshes for thermal-hydraulics, thermal-mechanics and neutron diffusion. For the same reason, an adaptive time step is employed. The different equations can be solved altogether or selectively included. The coupling follows a traditional merged PISO-SIMPLE loop for velocity, pressure and energy, while a more implicit coupling is achieved via

Picard iteration for the other fields. Parallel computations can be performed via domain decomposition.

In this paper, each “single-physics” sub-solver has first been presented, including basic information on the theoretical background and numerics, verification tests and examples. The coupling algorithm has then been described. Finally, a code-to-code benchmark with the system code TRACE has been performed. The GeN-Foam multi-physics solver has passed basic verification tests and compares well with TRACE solutions both at steady-state and for transient analysis. It shows good flexibility in terms of complexity of simulation (and computational requirements) thanks to the use of three unstructured meshes, the possibility to selectively include the equations to solve, and the possibility to treat complex structures using a coarse-mesh porous medium approach.

GeN-Foam is currently a complete multi-physics solver that can be used for the analysis of pin-based or homogeneous nuclear reactors. However, several assumptions and simplifications have been made throughout its development. In addition, optimization is still at a very early stage, and implementation of new features is foreseen to be necessary for specific applications and better predicting capabilities. Some of the weakest points currently being considered for future development work are reported here below:

- The compressible flow in porous media is a complex field that has been confronted here by introducing several assumptions such as: (1) a simplified $k-\varepsilon$ model; (2) neglect of the last term in Eq. (7); (3) a simplified treatment of the boundaries between zones with different porosities; (4) thermal equilibrium assumption for heat transfer correlations. In addition, several aspects would deserve further investigation such as general boundary effects in porous media, compressibility effects and numerical issues. At the current state of development, GeN-Foam is targeted mainly toward low Mach number flows.
- Inclusion of a two-phase porous medium model for thermal-hydraulics would be needed to allow e.g., for the simulation of severe accidents in SFRs, and for both operational and accidental transients in LWRs.
- The thermal-mechanics solver employed has limited applicability in cases where complex structures are modeled using a coarse mesh. This would require the development of an orthotropic thermal-mechanics sub-solver, as well as a rigorous coarse/fine-mesh approach for the thermal-mechanics of nuclear reactors.
- A strongly simplified fuel temperature model is currently employed. Inclusion of the fuel behavior code FRED (Mikityuk and Shestopalov, 2011) as a subroutine of OpenFOAM® to substitute or supplement the current model is being considered, together with coupling with the TRANSURANUS fuel behavior code.
- Application to different reactor systems and systematic validation or code-to-code benchmark would be needed to identify main modeling and numerical issues for different applications. Application to LWRs, SFRs and MSRs is currently envisioned at PSI.
- Possibility of analyzing LWRs has been mentioned based on the capability of GeN-Foam to model all necessary phenomena. However, such possibility would have to be confirmed via application to a real case to exclude major modeling or numerical issues. For instance, neutron production and consumption are strongly decoupled in thermal reactors and this may cause numerical oscillations in the multi-group neutronics sub-solver. In addition, code developments would be necessary such as implementation of water/steam properties, use of assembly power peaking factors, and inclusion of assembly discontinuity factors in the neutron diffusion sub-solver (which may hinder the use of high-order discretization schemes due to the presence of discontinuities).

- Good parallel computation performances have been observed for the cases considered in the paper, as expected from a standard use of the OpenFOAM® library. However, systematic scalability tests for different reactors and degrees of approximation should have to be performed to confirm these good performances.

Nomenclature

Latin symbols

$A_{fD,i}$	constant defined in Eq. (13) (–)
A_I	constant defined in Eq. (21) (–)
A_I	constant defined in Eq. (22) (–)
$A_{Nu,i}$	constant defined in Eq. (16) (–)
A_V	volumetric area between coolant and sub-scale structure ($\text{m}^2 \text{m}^{-3}$)
$B_{fD,i}$	constant defined in Eq. (13) (–)
B_I	constant defined in Eq. (21) (–)
$B_{Nu,i}$	constant defined in Eq. (16) (–)
C_k	concentration of the k th delayed neutron precursor group (m^{-3})
$C_{Nu,i}$	constant defined in Eq. (16) (–)
$C_{p,c}$	specific heat capacity of the cladding ($\text{J kg}^{-1} \text{K}^{-1}$)
$C_{p,f}$	specific heat capacity of the fuel ($\text{J kg}^{-1} \text{K}^{-1}$)
$C_{p,ss}$	specific heat capacity of the sub-scale structure ($\text{J kg}^{-1} \text{K}^{-1}$)
D	neutron diffusion coefficient (m)
D_f	fuel displacement field (m)
D_h	hydraulic diameter (or characteristic length) of the sub-scale structure (m)
$D_{Nu,i}$	constant defined in Eq. (16) (–)
e	coolant total energy (J kg^{-1})
E	Young modulus (MPa)
$f_{D,i}$	Darcy friction factor in the i th principal direction of the sub-scale structure (–)
$\mathbf{F}_g$	volumetric force due to gravity (N m^{-3})
$\mathbf{F}_{ss}$	volumetric force due to the interaction with the sub-scale structure (N m^{-3})
$F_{ss,i}$	component of $\mathbf{F}_{ss}$ in the i th principal direction of the sub-scale structure (N m^{-3})
h	heat transfer coefficient between coolant and sub-scale structure ($\text{W m}^{-2} \text{K}^{-1}$)
h_{eff}	heat transfer coefficient between the coolant and an external heat sink ($\text{W m}^{-2} \text{K}^{-1}$)
h_i	heat transfer coefficient in the i th principal direction of the sub-scale structure ($\text{W m}^{-2} \text{K}^{-1}$)
I	turbulent intensity (–)
k	turbulent kinetic energy ($\text{m}^2 \text{s}^{-2}$)
$\kappa(\mathbf{u}_D)$	tensor defined in Eq. (9) ($\text{N m}^{-4} \text{s}$)
$\kappa(u_D)_{ii}$	component of $\kappa(\mathbf{u}_D)$ in the i th principal direction of the sub-scale structure ($\text{N m}^{-4} \text{s}$)
k_c	conductivity of cladding ($\text{W m}^{-1} \text{K}^{-1}$)
k_{eff}	effective multiplication factor (–)
k_f	conductivity of fuel ($\text{W m}^{-1} \text{K}^{-1}$)
$\mathbf{k}_{ss}$	conductivity of the sub-scale structure ($\text{W m}^{-1} \text{K}^{-1}$)
k_T	turbulent conductivity ($\text{W m}^{-1} \text{K}^{-1}$)
k_0	target turbulent kinetic energy ($\text{m}^2 \text{s}^{-2}$)
l	turbulent length scale (m)
Nu_i	Nusselt number in the i th principal direction of the sub-scale structure (–)
p	pressure of the coolant (Pa)
p_i	pressure at the interface between coolant and sub-scale structure (Pa)
$\dot{Q}_f$	heat source in the fuel (W m^{-3})
$\dot{Q}_{ss}$	heat transferred from the sub-scale structure to the fluid (W m^{-3})

Re	Reynolds number for the fluid in the sub-scale structure (–)
S_d	delayed neutron source ($\text{m}^{-3} \text{s}^{-1}$)
$S_{n,i}$	fission neutron source from neutron energy groups others than the i th ($\text{m}^{-3} \text{s}^{-1}$)
$S_{s,i}$	scattering neutron source from neutron energy groups others than the i th ($\text{m}^{-3} \text{s}^{-1}$)
t	time (s)
T	temperature of the coolant (K)
T_c	temperature of the cladding (K)
T_{ext}	temperature of an external heat sink (K)
T_f	temperature of the fuel (K)
$T_{f/c}$	temperature in fuel or cladding (K)
$T_{f/c,ref}$	reference temperature in fuel or cladding (K)
T_{ss}	temperature of the sub-scale structure (K)
$\mathbf{u}$	coolant velocity (m s^{-1})
$\mathbf{u}_D$	Darcy coolant velocity (m s^{-1})
$u_{D,i}$	Darcy coolant velocity in the i th principal direction of the sub-scale structure (m s^{-1})
$\mathbf{v}_f$	versor in the axial fuel direction (–)
v_i	average neutron velocity for the i th energy group (m s^{-1})

Greek symbols

α	linear expansion coefficient of structures (K^{-1})
$\alpha_{f/c}$	linear expansion coefficient of fuel or cladding (K^{-1})
$\beta_{eff,t}$	total effective delayed neutron fraction (–)
$\beta_{eff,k}$	effective delayed neutron fraction for the k th delayed neutron precursor group (–)
$\beta_{0,t}$	total delayed neutron fraction (–)
γ	porosity (–)
ε	turbulent dissipation rate ($\text{m}^2 \text{s}^{-3}$)
ε_0	target turbulent dissipation rate ($\text{m}^2 \text{s}^{-3}$)
μ_T	turbulent kinematic viscosity (Pa s)
$\lambda_{\varepsilon/k}$	convergence rate of ε and k toward their target value (s^{-1})
λ_k	decay constant for the k th delayed neutron precursor group (s^{-1})
ν	average number of neutrons per fission (–)
ρ	density of the coolant (kg m^{-3})
ρ_c	density of the cladding (kg m^{-3})
ρ_f	density of the fuel (kg m^{-3})
ρ_{ss}	density of the sub-scale structure (kg m^{-3})
$\Sigma_{d,i}$	disappearance cross section for the i th energy group (m^{-1})
$\Sigma_{f,i}$	fission cross section for the i th energy group (m^{-1})
$\Sigma_{j \rightarrow i}$	group-transfer cross section from the j th to the i th energy group (m^{-1})
φ_i	neutron flux for the i th energy group ($\text{m}^{-2} \text{s}^{-1}$)
$\chi_{d,i}$	delayed neutron yield for the i th energy group (–)
$\chi_{p,i}$	prompt neutron yield for the i th energy group (–)

Acronyms

BC	boundary conditions
CASL	Consortium for Advanced Simulation of LWRs
CFD	computational fluid dynamics
ESFR	European sodium fast reactor
EURATOM	European Atomic Energy Community
FP7	seventh European Framework Program
FR	fast reactor
GeN-Foam	Generalized Nuclear Foam
LWR	light water reactor
MSR	molten salt reactor
PIMPLE	merged PISO-SIMPLE algorithm
PISO	pressure implicit split operator
PSI	Paul Scherrer Institut

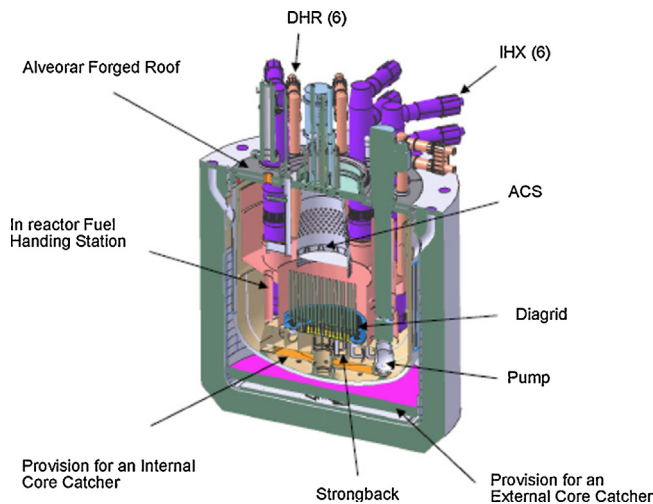


Fig. 11. ESFR pool-type concept (Fiorini and Vasile, 2011).

RANS	Reynolds averaged Navier–Stokes
SFR	sodium-cooled fast reactor
SIMPLE	semi-implicit method for pressure linked equations
ULOF	unprotected loss of flow
UTOP	unprotected transient overpower

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Appendix A. The European sodium fast reactor – ESFR

The ESFR (Figs. 11 and 12) is a 3.6 GW_{th} pool-type sodium-cooled FR. It has been designed during the EURATOM ESFR project within the seventh European Framework Program (FP7) and design details

Table 4
Nominal conditions and plant parameters used in the ESFR simulations (Fiorini and Vasile, 2011; Lázaro et al., 2014).

Reactor power (GW _{th})	3.6
Number of fuel assemblies	456
Assembly pitch (mm)	210.8
Wrapper flat to flat outer (mm)	206.3
Wrapper flat to flat inner (mm)	197.3
Number of pins per FA	271
Fuel pin outer clad diameter (mm)	10.73
Fuel pin inner clad diameter (mm)	9.73
Fuel pellet outer diameter (mm)	9.43
Fuel pellet inner diameter (mm)	2.4
Cladding material	ODS Steel
Fuel pellet material	(U,Pu)O ₂
Core inlet temperature (°C)	395
Core outlet temperature (°C)	545
Total mass flow rate through the active core (kg s ⁻¹)	18940

Table 5
Material properties used in the ESFR simulations (Fink and Leibowitz, 1996; MATPRO, 2003).

Sodium (743 K)	
Density (kg m ⁻³)	841.06
Specific heat capacity (J kg ⁻¹ K ⁻¹)	1267.13
Dynamic viscosity (Pa s)	2.54 × 10 ⁻⁴
Fuel	
Density (kg m ⁻³)	10480
Specific heat capacity (J kg ⁻¹ K ⁻¹)	250
Thermal conductivity (W m ⁻¹ K ⁻¹)	3
Linear expansion coefficient (K ⁻¹)	1.1 × 10 ⁻⁵
Cladding	
Density (kg m ⁻³)	7500
Specific heat capacity (J kg ⁻¹ K ⁻¹)	500
Thermal conductivity (W m ⁻¹ K ⁻¹)	20
Linear expansion coefficient (K ⁻¹)	1.8 × 10 ⁻⁵
Core structures (steel)	
Density (kg m ⁻³)	8000
Specific heat capacity (J kg ⁻¹ K ⁻¹)	600
Thermal conductivity (W m ⁻¹ K ⁻¹)	20
Linear expansion coefficient (K ⁻¹)	1.8 × 10 ⁻⁵
Gap conductance (W m ⁻² K ⁻¹)	3000

can be found in references Fiorini and Vasile (2011) and Lázaro et al. (2014). A slightly modified and simplified design has been used in this work that features a 60 degree symmetry in the core configuration (Fig. 12b). Tables 4 and 5 lists the main parameters and physical properties used for simulations in the present paper. It should be noted that constant physical properties are used, but the

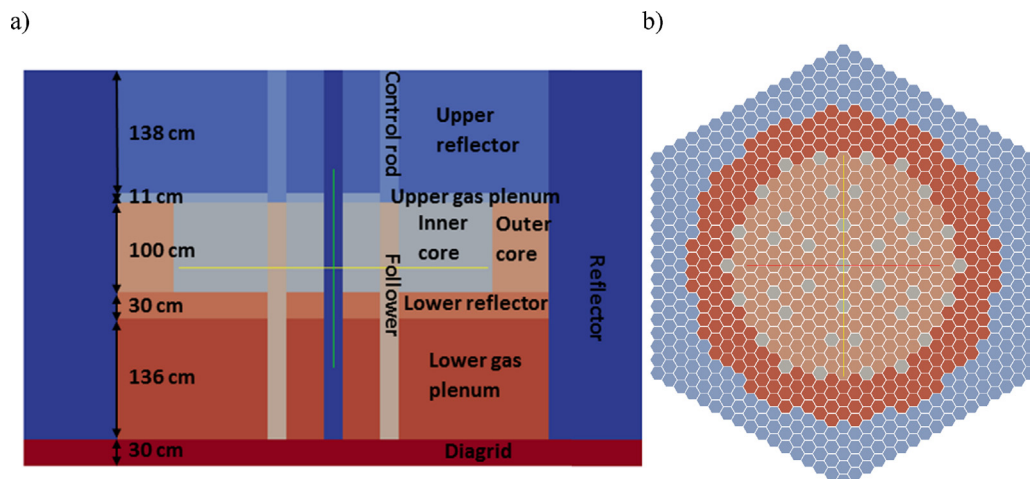


Fig. 12. Axial (a) and radial (b) configurations of the reference ESFR core.

possibility of temperature dependent properties is included in GeN-Foam. Pressure dependence is also allowed for coolant properties.

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